

Thanh Dang

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science, Computer Science

August 2022 – May 2026

GPA: 3.72

- **Honors:** Presidential and Dean's List: 2022-2026

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- **Frameworks:** React, React Native, Next.js, NestJS, Flask, Ray, PyTorch
- **Tools & Technologies:** Firebase, AWS, PostgreSQL, MySQL, Docker, Kafka, Redis, Node.js, Git

EXPERIENCE

Research Intern, **CPP Enterprises**, Pomona, CA

May 2025 – Present

- Developed GPU-enabled video processing service using Kafka for job ingestion and Ray to orchestrate multi-step, asynchronous pipelines, delivering real-time status updates via Server-Sent Events and Redis pub/sub
- Implemented NestJS REST API serving multiple features including route design, service-layer logic, and PostgreSQL persistence; deployed on Jetstream2 via CI/CD pipeline and Docker Compose orchestration
- Spearheaded development of audio-based task identification using WhisperX transcription with speaker diarization and Google Gemma LLM few-shot prompting, processing 50+ hours of mobility scooter driving footage

Undergraduate Research Assistant, **Cal Poly Pomona**, Pomona, CA

October 2024 – May 2025

- Built Python desktop application for Delta NCSA HPC cluster file management via SFTP, streamlining upload, download, and deletion workflows; presented the project at the CPP RSCA Conference to 20+ attendees
- Aligned label, motion, and pose data across 40 recording sessions by filtering non-driving segments, reducing dataset size by 15% while retaining 100% of relevant driving events for downstream analysis
- Trained self-supervised Transformer-based model to quantify upper-body asymmetry from 2D pose sequences, achieving strong separation between stroke and healthy groups using contrastive learning and ranking objectives

Code Coach, **theCoderSchool**, Cerritos, CA

January 2024 – February 2025

- Led coding classes for 30+ elementary students with no prior programming experience, teaching fundamental programming concepts and guiding students to build functional platformers and dialogue-driven games in Scratch
- Coached 5 students one-on-one in front-end web development, advancing from core HTML and CSS layout principles to designing and building interactive web pages with JavaScript, leveraging Replit for live deployment

PROJECTS

TrailTalk

September 2025 – December 2025

- Launched social media application for park enthusiasts using React and Firebase in Agile team environment, featuring 63 U.S. national parks with dedicated park pages, user reviews, photo sharing, and following features
- Designed scalable class-based NoSQL data model supporting user accounts, messaging, and content management, utilizing MapLibre for interactive mapping and geolocation features across 50 states
- Implemented live private messaging system with TypeScript and Cloud Firestore, incorporating notifications and read receipts to enable seamless user-to-user communication with low-latency message delivery

Carbon Closet, **Cal Poly Pomona Hackathon 2025**

April 2025

- Built full-stack e-commerce platform using Next.js and Firebase, enabling clothes buying, selling, and donations for sustainable fashion; presented to panel of 8 judges and secured 3rd place among 60 teams
- Developed responsive marketplace interface with Next.js, integrating Cloud Firestore for live listing updates, search filtering, shopping cart, user profile, and token-based reward and payment system
- Containerized application with Docker to ensure consistent environments across development and production; deployed to AWS EC2, gaining hands-on experience with cloud hosting and production deployment workflows

Recyclable or Not Bot, **UC Riverside Hackathon 2025**

January 2025

- Created interactive web application using React and Flask that classifies recyclable vs. non-recyclable items from camera input, promoting eco-friendly waste disposal; awarded best sustainability track among 50+ teams
- Integrated end-to-end pipeline using YOLOv8 deep learning model for real-time object detection, with Flask API managing OpenCV frame capture and delivering inference results to React front-end
- Developed interface with homepage displaying recycling tips and waste reduction resources, and camera view rendering real-time classification, color-coded recyclability score, and item labels

EXTRACURRICULARS

STEM Advantage Scholar: Scholarship program for first-generation college students in STEM

June 2024 – Present

Software Engineer Association: Largest software engineering community at CPP

November 2023 – May 2024